

Co2- AT1- Analytical
Problem.

Assessment

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CSA0307 Data Structures

Compare the Performance of a Linear Queue and a Circular Queue when 100 enqueue and dequeue operations are performed alternatively

when performing 100 Alternative enqueue and dequeue operations (i.e., Enqueue, dequeue, Enqueue, dequeue etc--)

→ Linear Queue Implementation

1) Using Array (without shifting elements):

- Behaviour:- Every Enqueue moves the Rear Index forward by 1, and Every Dequeue moves the front Index by one.
- Memory Failure:- Even though the queue never holds more than 1 element at any given time, after the 100th operation, The rear pointer reaches Index 100 (The end of the Array Allocated size)
- Result:- The queue will report "Queue full" Prematurely after N operations unless the Array size is at least 100.

→ using Array (with Shifting Elements on Dequeue)

- If elements are shifted left on every Dequeue to reuse index 0, every Dequeue takes $O(N)$ time complexity.

- Result: Time Performance drops significantly to $O(N)$ per Operation.

2) Circular Queue Implementation

Behaviour: when Rear or Front reaches the end of the Array, It wraps around to the beginning (0) using Modulo Operator (Arithmetic).

Result: Both Enqueue and dequeue run in Constant time $O(1)$.

Space efficiency: An Array Size of Just 2 elements is sufficient to perform all the Alternate Operations Seamlessly, As freed spaces at Index 0 are Immediately reused.

Deque Queue
Full" even though some positions are Empty
Implemented using Array reports "Queue

- Deque
- Analyze why this occurs.
 - Explain How a Circular queue solves this Problem.

- In standard linear queue Implemented using a fixed-size Array.
- Elements are Inserted at the Rear Index and deleted from the Front Index.
- when an element is dequeued, the Front Index Increments, leaving the previous Index Empty.
- The condition used to check if the queue is full is Typically:
$$\text{Rear} == \text{Capacity} - 1$$

The Problem: → The rear pointer only moves forward and never moves backward. once rear reaches the final Index of the Array the condition returns True, even if all were

Emptied by previous dequeue operations.

b) A circular queue treats the Array as a ring or continuous loop where the position following the last index (capacity - 1) is index 0.

Key Mechanics:-

1) Modulo Arithmetic for Indexing:-

Instead of simple incrementing ($R = R + 1$), indices wrap around modulo division.

2) Reusing Empty slots:- when Rear reaches the end of the Array, the Modulo operation redirects Rear back to index 0.

3) Updated full condition:- The queue is considered full only when the next position of rear matches $\text{Front} = (\text{Rear} + 1) \pmod{\text{capacity}} = \text{Front}$.